

Reading: Examples to ALTER and TRUNCATE tables using MySQL

Estimated time to complete: 5 minutes

In the previous video, the ALTER and TRUNCATE syntax applies to DB2. There are variations in syntax between different databases. This reading will explore some examples of ALTER and TRUNCATE statements using MySQL.

Objective(s)

At the end of this reading, you will be able to:

- Use the ALTER TABLE statement in the correct syntax.
- Use TRUNCATE statements in syntax.
- Execute examples of ALTER and TRUNCATE statements.

ALTER TABLE

ALTER TABLE statements can be used to add or remove columns from a table, to modify the data type of columns, to add or remove keys, and to add or remove constraints. The syntax of the ALTER TABLE statement is:

ADD COLUMN syntax

```
ALTER TABLE table_name
ADD column_name data_type;
```

A variation of the syntax for adding column is:

```
ALTER TABLE table_name
ADD (column_name, column_name, data_type);
```

By default, all the entries are initially assigned the value null. You can then use insert statements to add the necessary column values.

For example, to add a telephone_number column to the author table in the library database, the statement will be written as:

```
ALTER TABLE author
ADD telephone_number CHAR(20);
```

Here, BIGINT is a data type for big integer.

After adding the entries to the new column, a sample output is shown below.

author_id	lastname	firstname	email	city	country	telephonumber
1001	Thomas	John	john@...	New York	USA	5551111
1002	James	Alice	alicej@...	Seattle	USA	5551112
1003	Wells	Steve	steve@...	Montreal	Canada	5552222
1004	Kumar	Santosh	kumars@...	London	UK	5553333

Modify column data type

```
ALTER TABLE table_name
MODIFY column_name data_type;
```

Sometimes, the data presented may be in a different format than required. In such a case, we need to modify the data type of the column. For example, using a numeric data type for telephone_number means you cannot include parentheses, plus signs, or dashes as part of the number. For such entries, the appropriate choice of data type is CHAR.

To modify the data type, the statement will be written as:

```
ALTER TABLE author
MODIFY telephone_number CHAR(20);
```

The entries can then be updated using UPDATE statements. An updated version of the "author" table is shown below.

author_id	lastname	firstname	email	city	country	telephonumber
1001	Thomas	John	john@...	New York	USA	5551111
1002	James	Alice	alicej@...	Seattle	USA	5551112
1003	Wells	Steve	steve@...	Montreal	Canada	5552222
1004	Kumar	Santosh	kumars@...	London	UK	5553333

TRUNCATE Table

TRUNCATE TABLE statements are used to delete all of the rows in a table. The syntax of the statement is:

```
TRUNCATE TABLE table_name;
```

So, to truncate the "author" table, the statement will be written as:

```
TRUNCATE TABLE author;
```

The output would be as shown in the image below.

author_id	lastname	firstname	email	city	country

Note: The TRUNCATE statement will delete the rows and not the table.

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